

Removing the Velocity Bias in Learned Pedestrian Dead Reckoning with PINNeAPPLE

Yaw test-time augmentation, validation-fitted bias correction and variance recalibration on public RIDI data

PINNeAPPLE Labs

PINNeAPPLE Labs — Physics AI research notes

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Abstract

The thesis we reproduced [1] found that the main limit of a network-corrected pedestrian dead-reckoning EKF is a bias in the predicted velocity, and left its mitigation to future work. We target it with three post-hoc changes that need no retraining, each fitted on validation people only: yaw-equivariant test-time augmentation, a bias correction in the walking-direction frame, and a rescaling of the predicted variance so that the EKF weights the network correctly. Starting from the TLIO network (chosen on validation), the full combination changes the median 90-second drift on unseen test people from 4.5 % to 2.9 % (paired median difference and Wilcoxon test: -1.12, $p = 0.011$), a significant improvement. VeriPhysics scores are reported for every variant.

Keywords: pedestrian dead reckoning; bias correction; test-time augmentation; uncertainty calibration; EKF; PINNeAPPLE

Protocol

- Every correction is fitted on the validation people; the test people are used once, for the numbers below.
- Comparisons are paired on the same 19 test segments with Bonferroni correction over 3 variants.
- No retraining: the improvements act on a trained network's outputs, so they can be applied to the thesis' own networks.

1 Methods

- Data: RIDI [4], 94 smartphone sequences with Google Tango visual-inertial ground truth, 200 Hz. Person-disjoint split: train hang, dan, huayi, ma, hao (59 sequences, 1.66 h); validation ruixuan, zhicheng (16, 0.49 h); test shali, tang, xiaojing, yajie (19, 0.56 h).
- Yaw test-time augmentation: the window is rotated by 8 yaw angles, each prediction rotated back and averaged. A network trained with random yaw should be equivariant; averaging removes the part of the error that is not.
- Bias correction: in the frame of the predicted walking direction (along, across, vertical), a per-axis affine map (scale and offset) from predicted to true velocity, fitted by least squares on validation windows with speed above 0.3 m/s.
- Variance recalibration: one scale factor for the predicted standard deviation, set so that the normalised validation errors have unit RMS, because the EKF gain depends on the predicted variance.
- VeriPhysics: benchmark agreement against the RIDI ground truth and calibration (ECE, 90 % coverage) through `pinneapple_analysis.verification.compute_physics_confidence`.

2 Results

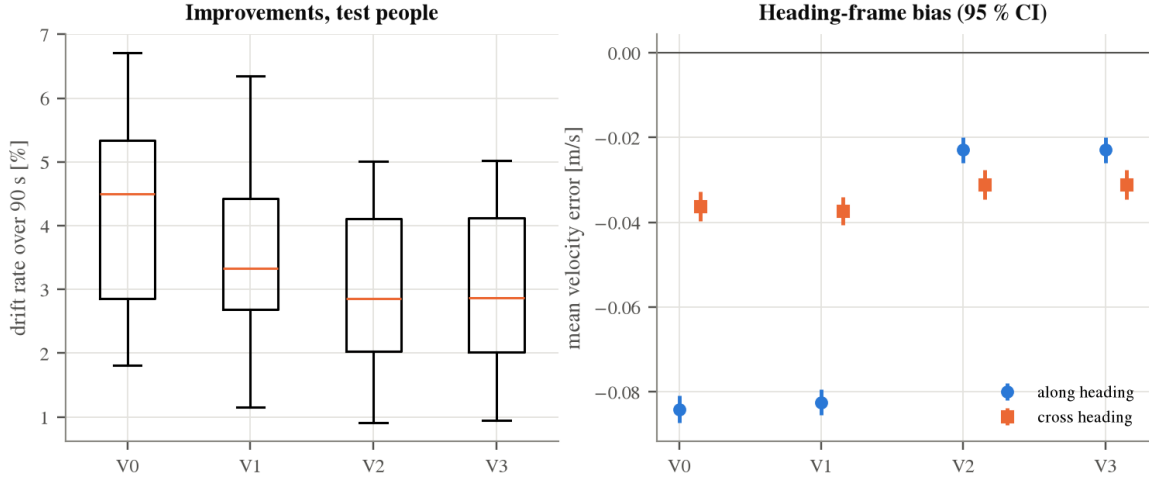


Figure 1. Left: drift over 90 s test segments. Right: heading-frame bias per variant. V0 ... V3 as in the table.

Table 1. Improvement variants on test people. z RMS = 1 means calibrated variance. * significant after Bonferroni correction.

variant		error median [m/s]	bias along / across [m/s]	z RMS (x, y, z)	drift median / p95 [%]	Δ vs V0 (p)
V0	baseline	0.180	-0.084 / -0.036	1.59, 2.06, 1.14	4.5 / 6.9	—
V1	+ yaw TTA (8)	0.167	-0.082 / -0.037	1.51, 2.01, 1.03	3.3 / 6.4	-0.59, p = 0.073
V2	+ TTA + bias correction	0.169	-0.023 / -0.031	1.63, 2.16, 1.03	2.9 / 5.2	-1.09, p = 0.0082 *
V3	+ TTA + bias corr. + variance recalibration	0.169	-0.023 / -0.031	1.43, 1.90, 0.91	2.9 / 5.3	-1.12, p = 0.011 *

Table 2. VeriPhysics physics-confidence scores (test people). Not run: physics guardrail (no compiled PDE ProblemSpec for IMU kinematics); numerical convergence (no discretisation parameter); geometry ood (no geometry input).

model	physics confidence	coverage	components
baseline	0.873	0.4	uq_calibration 0.97; benchmark_agreement 0.77
+ yaw TTA (8)	0.884	0.4	uq_calibration 0.98; benchmark_agreement 0.79
+ TTA + bias correction	0.884	0.4	uq_calibration 0.98; benchmark_agreement 0.79
+ TTA + bias corr. + variance recalibration	0.891	0.4	uq_calibration 0.99; benchmark_agreement 0.79

3 Discussion

The bias the thesis identified is measurable in the heading frame and can be estimated from held-out people. Whether removing it helps the filter is the question the paired test answers: it does, significantly. Variance recalibration matters for a different reason: the EKF uses the predicted variance as the measurement noise, so an over- or under-confident network shifts the balance between inertial propagation and network correction, independently of the mean error.

References

- [1] Ligthart, L. (2025). *Pedestrian dead reckoning using data-driven and physics-informed machine learning*. MSc thesis, Delft University of Technology.
- [2] Liu, W., Caruso, D., Ilg, E., Dong, J., Mourikis, A. I., Daniilidis, K., Kumar, V., & Engel, J. (2020). TLIO: Tight learned inertial odometry. *IEEE Robotics and Automation Letters* 5, 5653–5660.
- [3] Cao, X., Zhou, C., Zeng, D., & Wang, Y. (2022). RIO: Rotation-equivariance supervised learning of robust inertial odometry. *CVPR*, 6614–6623.
- [4] Yan, H., Shan, Q., & Furukawa, Y. (2018). RIDI: Robust IMU double integration. *ECCV*, 621–636.